



# 1977 - Glows in the Dark: New Models for the Atmospheric Structure and Evolution of High Metallicity and Low Mass Giant Planets

Cycle: 1, Proposal Category: AR

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## OBSERVATIONS

### ABSTRACT

JWST is expected to discover a host of young, self-luminous, low mass giants planets in the coming years. The telescope will be capable of detecting planets comparable to our own Uranus and Neptune when they were young and glowing brightly in the thermal infrared. The atmospheric structure and evolution models that are needed to interpret these planets and place them in context, however, do not currently exist or are not adequate for the job of characterizing these expected discoveries. We propose to develop grids of exoplanet atmospheric structure, spectra, and evolution models to lower masses and higher metallicities than have yet been computed. Such models will facilitate the planning and interpretation of future observations of cool, low mass directly imaged planets by JWST. We will deliver our grids and the infrastructure needed to understand and manipulate them to the community as a resource to support the study of young giant planets into the future.

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